Band-1 Hadamard Order-2036 Multi-Method Lit-Watch / Eligibility Triage

Miyamoto existence reduction to H_508; H_2036 UNCONSTRUCTED on box; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / literature ledger increment (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-14_Band1_Hadamard_2004_lit_watch`; `2026-09-14_Band1_Hadamard_le64_complete_incre`
Lab. `/workspace/experiments/band1/hadamard_2036_method_triage/`.
Cutoff. 2026-09-14.

Honesty stamp (non-negotiable).
This memo is a **multi-method eligibility triage** for order **2036**. It **does not** construct an explicit ± 1 matrix of order 2036 on box. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. `clay_closed = false`.

Abstract

After the Band-1 correction that order **2004** is classical Paley type I ($q = 2003$), the first gap in the Sylvester∪Paley∪Kronecker closure of {all ≤ 2000}∪Paley is **2036**. This increment triages standard methods for that gap. Paley I/II and nontrivial Kronecker all **fail**. Miyamoto (1991) **applies as an existence reduction**: $q = 2036/4 = 509$ is prime with $509 \equiv 1 \pmod 4$, and H_{508} is literature-known after the August 2026 fill of all admissible orders ≤ 2000. On this box, however, no H_{508} payload is present, so H_{2036} **remains UNCONSTRUCTED**. Unit tests **14/14 PASS**. Among the first 24 standard-closure gaps, Miyamoto-eligible orders are **2036, 2116, 2164, 2228, 2276**. The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 2036; Miyamoto construction; method triage; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemma (this increment). Method eligibility classification for $n = 2036$, plus a Miyamoto **existence reduction** to literature H_{508}. **Not** an on-box construction. **Not** the Hadamard conjecture.

2. Context after the 2004 errata

Prior increment: H_{2004} constructed and verified via Paley I ($q = 2003$). Popular secondary sources (Cook 2026-08-13; some Aug 2026 summaries) still name 2004 as the next open admissible order — that is **errata**.

Under Sylvester ∪ Paley I/II ∪ Kronecker closure of {all ≤ 2000}∪Paley (limit 4000), the first missing admissible order is **2036**.

3. Methods triaged

Method	Criterion	Result for 2036
Paley I	$q = n - 1$ prime power $\equiv 3 \pmod 4$	**FAIL** ($2035 = 5 \cdot 11 \cdot 37$)
Paley II	$q = n/2 - 1$ prime power $\equiv 1 \pmod 4$	**FAIL** ($q = 1017 = 3^2 \cdot 113$)
Kronecker	nontrivial factorization into two Hadamard-admissible orders	**FAIL** (admissible divisors only $\{1, 2, 4\}$)

Method	Criterion	Result for 2036
Miyamoto (1991)	$q = n/4$ prime power $\equiv 1 \pmod 4$ and H_{q-1} exists	**EXISTENCE OK** ($q = 509$ prime, $509 \equiv 1 \pmod 4$, H_{508} lit-known ≤ 2000)
Yamada (a)	$q \equiv 1 \pmod 8$ + $H_{(q-1)/2}$	**FAIL** ($509 \equiv 5 \pmod 8$)
On-box explicit ± 1 matrix	requires H_{508} payload	**UNCONSTRUCTED**

Miyamoto theorem (cited). If q is a prime power with $q \equiv 1 \pmod 4$ and a Hadamard matrix of order $q - 1$ exists, then a Hadamard matrix of order $4q$ exists (Miyamoto, *J. Combin. Theory Ser. A* **57** (1991) 86–108).

Caveat. H_{508} itself is outside the Sylvester∪Paley∪Kronecker seed closure (factors of 508 yield no nontrivial Hadamard split). Its literature existence after August 2026 does **not** put an explicit payload on this box.

4. Unit tests

Test	Result
H_{20} Paley I control	PASS
H_4 Sylvester	PASS
H_{12} Paley I	PASS
Miyamoto predicate $n = 20$ eligible	PASS
Miyamoto predicate $n = 24$ ineligible	PASS
2036 divisible by 4	PASS
509 prime	PASS
$509 \equiv 1 \pmod 4$	PASS
508 Hadamard-admissible	PASS
Paley I/II/Kronecker fail for 2036	PASS
Miyamoto eligible for 2036	PASS
Reject all-ones 4×4	PASS

All 14/14 unit tests PASS.

5. Gap scan (first 24 standard-closure misses)

Miyamoto-eligible among first 24 gaps after 2000:

- **2036** ($q = 509$ prime)
- **2116** ($q = 529 = 23^2$)
- **2164** ($q = 541$ prime)

- **2228** ($q = 557$ prime)
- **2276** ($q = 569$ prime)

None of these is claimed constructed on box this increment. Each is an existence-reduction candidate contingent on the matching H_{q-1} payload.

6. Literature watch notes

Source	Relevance
Wikipedia *Hadamard matrix* (fetched context 2026-09-14)	Aug 2026 Alpöge et al. filled twelve gaps ≤ 2000; conjecture OPEN
Cook blog 2026-08-13	Still cites 2004 as not known in S — **conflicts** with Paley I (prior errata)
Miyamoto 1991	Direct existence path $H_{508} \Rightarrow H_{2036}$
SageMath Hadamard docs	Names Miyamoto among implemented methods (coverage primarily ≤ 1208 + some larger)

August 2026 fill (background). Orders 668, 716, 892, 1132, 1244, 1388, 1436, 1676, 1772, 1916, 1948, 1964 announced; implies every admissible ≤ 2000 is literature-known, including **508**.

7. Non-claims

- Full Hadamard conjecture: **OPEN** (Band-3).
- H_{2036} on box: **UNCONSTRUCTED** (existence-reduced only).
- No Clay Millennium problem closed; `clay_closed = false`.
- GATE-04 collection: **LOCKED** (no invented J).
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- No claim that Williamson / Goethals–Seidel / Turyn / other searches have been exhausted for 2036 worldwide.

8. Reproducibility

```
cd /workspace/experiments/band1/hadamard_2036_method_triage
python3 verify_hadamard_2036.py
```

Emits `results.json`, `literature_ledger.json`, `SUMMARY.txt`.

9. Conclusion

Outcome. Order **2036** is the first standard-closure gap after the 2004 Paley-I correction. It is **not** Paley/Kronecker-constructible from the ≤ 2000 seed set. It **is** Miyamoto-eligible as an existence

reduction to literature H_{508}. Explicit on-box construction awaits an H_{508} (or other) payload. Full conjecture remains **OPEN**.

Next optional. Ingest/construct H_{508} then run Miyamoto for H_{2036}; or triage 2052/2060 by other families; or Track B privacy / Object 1 meter await.

Shelf. `2026-09-14_Band1_Hadamard_2036_method_triage.md` / `.pdf` under `/workspace/DOWNLOAD_PDFs/`.